

PROVEIT RESEARCH COMPANION

Specialization-Safe Radical Solvers

Exact Bad Characteristics for Subset-Sum Resolvents,
a Fourier–Kummer Atlas, and a Global Obstruction

Research prepared for Vladimir Reshetnikov

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ProveIt research-report collection, continuing the formal project `Algebra/PolynomialFormulas`. Batch 36, manuscript 04 (archive `ProveIt_Radical_Solvers_Research`); pinned to ProveIt `e21766d04`. AI-assisted, unrefereed, not formalized. Notes marked `[write]` were added when the manuscript was written into the collection; see Section 1.5.

Abstract

A separable polynomial can have an inseparable auxiliary resolvent; a correct radical identity can become unusable on a nonempty exceptional locus; and separately chosen radical branches need not reconstruct one compatible root tuple. We study these three problems in the prime-degree setting of the ProveIt polynomial-formula project.

For equal-cardinality subset sums of the roots of an irreducible prime-degree polynomial, we give an exact, finite cyclotomic-resultant criterion for the characteristics in which collisions are possible. It yields the complete exceptional sets $\{5\}$ for quintic pair sums, $\{2, 7\}$ for septic pair sums, and $\{2, 7, 29\}$ for septic triple sums. In particular, over $\mathbb{F}_{29}(t)$ the separable irreducible polynomial $X^7 - t$ has triple-sum resolvent

$$(Y^7 - 12t)^2(Y^7 - t)(Y^7 - 17t)(Y^7 - 28t).$$

In characteristic 29, collisions occur precisely for translated pure septics; in characteristic two they characterize quotients of additive degree-eight polynomials, up to translation. The explicit example supplies a concrete obstruction to an unrestricted distinct-factor reading of a published septic factor-pattern assertion. A polynomial separating-invariant construction repairs the specialization problem over any sufficiently large field; for septic triples, 1191 distinct parameter values always suffice.

We then prove a branch-coherent Fourier–Kummer reconstruction using one p th root on each nonvanishing Fourier chart. Every nonempty Fourier support occurs for an irreducible rational polynomial with full affine Galois group. Finally, the cyclic quotient of the ordered distinct-root configuration space has Picard group $\mathbb{Z}/p\mathbb{Z}$, and its universal cyclic cover has no global regular Kummer generator. Thus local formulas are not merely an artifact of a poor choice of coordinates. Complete arguments, finite integer certificates, reproducible exact checks, a repository integration plan, and further research questions are included.

Status. This is a mathematical research article with exact computational certificates, not a newly kernel-checked Lean development. The basic Fourier, Galois, resultant, and descent tools are classical. The specific classifications, examples, and synthesis below are proved here; priority over all existing literature has not been established. No claim to have solved the general problem of finding new radical formulas in every degree is made.

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1 Research target, provenance, and contribution boundary

1.1 Why this problem arises in ProveIt

The repository snapshot used for the investigation is commit

e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9.

The root README identifies the audit of Lazard’s solvable-quintic algorithm as an active direction. The polynomial-formula report distinguishes raw invariant equations from root-origin or Fourier evidence, and explicitly records exceptional denominators and the need for coherent radical choices [4, 5, 6]. These are mathematical issues, not only proof-assistant engineering issues.

The inspected source `LazardGeneralFourier.lean` already proves arbitrary-degree character orthogonality, cyclic-shift covariance, and Fourier inversion. It deliberately stops before branch-sensitive recovery of modes from invariants [7]. Accordingly, ordinary discrete Fourier inversion is not presented below as a new result. Our reconstruction uses that existing interface and adds a pivot chart, invariant ratios, and a proof that every choice of the one remaining radical changes only the cyclic ordering of the roots.

[write] **Repository context the manuscript did not cite.** Checked against the tree, which is unchanged in `Algebra/PolynomialFormulas` between the pin and batch 36.

- *The Rocq twin.* Beside the Lean module `Algebra/PolynomialFormulas/Lean/PolynomialFormulas/LazardGeneralFourier.lean` (`character_orthogonality`, l. 49; `dft_shift`, l. 57; `invDFT_dft`, l. 98; the scope boundary quoted above is its docstring, l. 18–21) the project has `Algebra/PolynomialFormulas/Coq/LazardGeneralFourier.v`, proving the same arbitrary-degree core over any MathComp field: `lazard_general_character_orthogonality` (l. 96), `lazard_general_dft_shift` (l. 128) and `lazard_general_invDFT_dft` (l. 207), with the coordinatewise `lazard_general_invDFT_dft_coordinate` (l. 179) [8]. The claim crosswalk (`Algebra/PolynomialFormulas/LazardPaperClaimCrosswalk.md`, l. 199) records both focused modules as kernel-green with their public aggregate/audit pending, and states the same boundary: arbitrary-degree branch-sensitive invariant recovery is deliberately outside the general Fourier modules, while the quintic invariant recovery is covered by a separate five-coordinate chain.
- *Lazard’s own one-radical scheme.* The pivot chart and invariant ratios are not new in the quintic case. The project report `Algebra/PolynomialFormulas/LazardQuinticFormalization.tex` (l. 606–612) records Lazard’s invariants $S_1 = s_1^5$, $S_2 = s_2 s_1^3$, $S_3 = s_3 s_1^2$, $S_4 = s_4 s_1$ and his reconstruction “once $P_1^5 = S_1$ is chosen with $P_1 \neq 0$ ”, with $P_2 = S_2/P_1^3$, $P_3 = S_3/P_1^2$, $P_4 = S_4/P_1$ [5]. This is the case $p = 5$, pivot $j = 1$ of Theorem 8.1: with $\zeta = \omega^{-1}$ one has $u_k = s_k$, $a_1 = S_1$, $c_{1,k} a_1 = S_k$ and $\rho = P_1$ (dictionary in Section 1.4). What the chart theorem adds is the arbitrary odd prime p , the arbitrary pivot j , the statement that *every* branch of the one radical yields the cyclic shift x_{i+rj-1} , the transition rules on overlaps, and Corollary 8.2. A name search of the Lean quintic modules at the write (for `shift` and `rotate`) found no theorem that every fifth-root choice gives a cyclic shift; that search is not a proof of absence.

Three precise research problems guide the article. First, determine exactly when a standard subset-sum resolvent can fail to separate subsets despite separability and irreducibility of the original polynomial. Second, replace an unsafe invariant by one with a proved specialization guarantee. Third, understand whether one can eliminate exceptional radical charts globally, or whether there is a structural obstruction. These are the questions formulated and answered here; they are not attributed to an existing issue or to a named open-problem list.

1.2 Relation to the published formulas

Lazard’s 2004 chapter gives explicit formulas for solvable quintics [1]. Faggal and Lazard’s 2014 article also treats septics, using sums of three roots and their factorization patterns; its specialized septic construction assumes characteristic different from 2, 3, 7 [2]. Thus the existence of a radical treatment of solvable septics is not a new claim of this article.

Our characteristic-29 example concerns the distinction between an orbit product and a squarefree resolvent with distinct conjugate values. It does not, on its own, disprove an abstract radical-solvability theorem. In particular, we do not infer that every later identity in that article fails. Section 6 states the precise implication and its limits.

1.3 What is established, and by what kind of evidence

Result	Mathematical content	Evidence here
Exact bad-characteristic criterion	Necessary and sufficient cyclotomic norm condition, with irreducible witnesses	General proof
Small-degree classification	Complete exceptional sets for $(p, k) = (5, 2), (7, 2), (7, 3)$	Proof plus finite integer tables
Characteristic-29 witness	Explicit repeated-factor triple-sum resolvent	Direct field calculation
Septic collision rigidity	Complete root-configuration classification in characteristics 2 and 29	General proof and finite norm certificate
Separating repair	Characteristic-independent subset descriptors and a finite parameter bound	General proof
One-radical charts	Coherent reconstruction and transition functions	General proof
Support realization	Every nonempty support with full affine Galois group	Explicit rational construction
Global obstruction	$\text{Pic}(A^{C_p}) \cong \mathbb{Z}/p\mathbb{Z}$ and no eigenunit	Algebraic descent and unit calculation
Lean integration	Proposed declarations and dependencies	Plan only; no new Lean build

The universal theorems do not depend on numerical experiments. The three small-degree classifications use a fully explicit finite calculation: all 850 unordered pairs are checked by exact resultants, and the 29 affine-orbit representatives are checked independently by integer determinants. The provided standard-library auditor checks the finite certificates without SymPy. This is still an ordinary computational proof boundary, not a proof-assistant kernel boundary.

1.4 [write] Conventions and notation

No symbol of the delivered manuscript was renamed and no normalization was changed. Several letters carry more than one meaning, inside the article or between the article and the formal project it continues; each is unambiguous in context, and the table fixes the readings.

Symbol	Meaning here	Not to be confused with
u_j, ζ	Fourier mode $u_j = \sum_i \zeta^{-ij} x_i$ (negative exponent), (14)	Lazard's $s_k = \sum_j \omega^{kj} x_j$ (positive exponent), in the project report and in <code>LazardGeneralFourier.lean</code> , whose character is called ψ . <i>True</i> : with $\zeta = \omega^{-1}$, $u_k = s_k$. <i>False reading</i> : $u_k = s_k$ for $\zeta = \omega$; then $u_k = s_{p-k}$.
$a_j, c_{j,k}, \rho$	Pivot power u_j^p , invariant ratios $u_k/u_j^{e_j(k)}$, a root of $T^p - a_j$ (Theorem 8.1)	For $p = 5, j = 1$: $a_1 = S_1$, $c_{1,k}a_1 = S_k$, $\rho = P_1$ in Lazard's notation [5]; $c_{1,k}$ itself is <i>not</i> S_k .
$e_j(k)$	The integer in $\{1, \dots, p-1\}$ with $je_j(k) \equiv k$ (Section 8)	e_r , the r th elementary symmetric function (Sections 5, 7).
$\mathcal{B}_{p,k}$	Set of bad primes (Section 3)	$B_{n,k} = (k-1)\binom{N}{2}$, an integer bound (11).
A, B	Subsets of indices (Sections 2–7)	B : the fixed field $E^{\langle \sigma \rangle}$ (Section 8), B_0, B in Section 9; $A, B = A^{C_p}$: rings in Section 12.
E	A field (Sections 8, 9); $E(A)$: the descriptor vector (Section 7)	Lazard's invariant denominator E in the project report.
k	Subset size (Sections 2–7); mode index in $c_{j,k}$; algebraically closed field (Section 12)	—
N	Norm $N_{A,B}$ (3); $N = \binom{n}{k}$ (Section 7); a minimal normal subgroup (Lemma 9.1)	—
f, g, m	The polynomial; but $f = e_h(k)$, $g = e_j(k)$ and $m = (ef - g)/p$ in (19)	m_{ij} , entries of the multiplication matrix M_q .
T	Polynomial variable (Sections 3, 4), Kummer variable in $T^p - a_j$	Lazard's quadratic-stage quantity T in the project report.
ψ	The $\mathbb{F}_2$ -linear map in the proof of Theorem 11.2	The additive character ψ of <code>LazardGeneralFourier.lean</code> .
L	Splitting field; L_W additive root polynomial; L_r character eigenspace modules (Section 12)	—
Δ	Discriminant $\prod_{i < j} (x_i - x_j)^2$ (25)	—
c	$c_r(A)$ (Theorem 5.1); $c_{j,k}$; c_k (Theorem 10.1); coefficient vector (Section 3); a constant in (27)	—

1.5 [write] Provenance of this report

This report is built from **one** manuscript: batch 36, manuscript 04 of ProveIt's incoming-reports intake, delivered as the archive `ProveIt_Radical_Solvers_Research` in commit `e13affd32`, pinned to ProveIt commit `e21766d04c2b8a9b2cd43e563024b3bd1b9`, and placed in commit `1a1396d4d`. It is filed in ProveIt's research-report collection and continues the formal project `Algebra/PolynomialFormulas`; it shares no theorem with any other report. Every statement, proof, table, example, remark, question and limitation of the delivered text is printed. No merge choice arose.

The write changed only the following. Every label carries the prefix `rss:`. The title-page provenance line, Sections 1.4, 1.5 and 1.6, the unnumbered [write] notes, and the bibliography entries [8], [9], [10] and [11] were added. The notes use no theorem counter, so every theorem, equation and table number is that of the delivered 29-page PDF. The delivered text names the delivery layout: its files `certificates/X` are shipped as `data/X`, its two scripts as `code/verify_results.py` and `code/audit_certificates.py`, and its `requirements.txt` as `data/requirements.txt`; the delivered PDF and README are not shipped. Appendix A gives commands that work with the shipped names.

1.6 [write] Formal status and what is not claimed

- **Not formalized.** No statement of this report is formalized in Lean or Rocq, and no `rss` label has a Lean or Rocq counterpart. That the report continues the `Algebra/PolynomialFormulas` development confers no formal status on it. The project's own nearby declarations are named where they are used: the arbitrary-degree Fourier core in Lean and Rocq (Section 1), sextic descriptor separation (Section 7), the cyclic Kummer generator (Section 8) and two positive-characteristic counterexamples (Section 3). None of them states a theorem of this report.
- **Integration plan.** Section 13 is a plan. None of it is formalized, and no Lean or Rocq code is delivered or shipped; its declaration names are proposals.
- **Evidence.** The general theorems rest on written proofs. The finite tables rest on exact computations by an ordinary Python program and an independent standard-library auditor, not a proof-assistant kernel. The Fourier checks cover coefficient-one supports only. Nothing computational checks the Picard-group and eigenunit section (Section 12), the transition rule (19), the Galois groups $\text{AGL}_1(\mathbb{F}_p)$, or the rigidity theorems beyond their finite character-pattern counts.
- **The source's own non-claims** stand as printed: the Status paragraph after the abstract (not a kernel-checked development; classical tools are not new; priority over the literature not established; no general radical formulas in every degree); the limits of Section 6 (distinct-factor separation only, not a complete audit of the published algorithm); the assumed invariant data of Theorem 8.1 (Section 8); relative minimality only in Corollary 8.2; minimality of the coordinate-pivot cover only in Corollary 10.2; what Theorem 12.1 does not rule out; and the status of the research questions (Section 14).
- **Review.** The report is AI-assisted and unrefereed. At the intake the proofs were read, the headline computations were repeated by a separate script, and both delivered suites were rerun on a copy; this is not an independent proof review. Parts of Section 3 (the relation-module viewpoint and characteristic-zero separation) may be close to the classical literature on linear relations among conjugates; this was not checked, and no priority is claimed.

2 Subset-sum resolvents and the correct notion of separation

Let p be an odd prime and let $f \in K[X]$ be monic, separable, and irreducible of degree p . In its splitting field choose a numbering $x_0, \dots, x_{p-1}$ of its roots. For $1 \leq k \leq p-1$ put

$$\mathcal{S}_{p,k} = \{A \subseteq \{0, \dots, p-1\} : |A| = k\}, \quad s_A = \sum_{i \in A} x_i,$$

and define the *orbit product*

$$R_{p,k,f}(Y) = \prod_{A \in \mathcal{S}_{p,k}} (Y - s_A). \quad (1)$$

Galois automorphisms permute the subsets, so this monic polynomial belongs to $K[Y]$. Its degree, with multiplicity, is always $\binom{p}{k}$.

Definition 2.1. The k -subset sums of f *separate* if $s_A \neq s_B$ for every two distinct members of $\mathcal{S}_{p,k}$. Equivalently, $R_{p,k,f}$ is separable, or $\gcd(R_{p,k,f}, R'_{p,k,f}) = 1$.

This is stronger than separability of f . The latter distinguishes the individual x_i , not all sums made from them. It is also important not to call (1) the minimal polynomial of a single value

without qualification. If the Galois group has several orbits on subsets, then even a separating orbit product is reducible. If distinct subsets have the same value, the product has repeated factors.

Lemma 2.2 (Orbit-to-factor principle). *Let a finite Galois group G act on a finite set $\mathcal{S}$, and let $A \mapsto v_A$ be a G -equivariant map into the Galois extension. If this map is injective, the irreducible factors over the fixed field of $\prod_{A \in \mathcal{S}} (Y - v_A)$ are indexed by the G -orbits on $\mathcal{S}$, and their degrees equal the orbit sizes.*

Proof. The conjugates of v_A are exactly v_{gA} for $g \in G$. Injectivity identifies its stabilizer with that of A . Its minimal polynomial is therefore the product over the orbit of A , with distinct roots. Different orbits have disjoint value sets, and multiplication gives the assertion. $\square$

Without injectivity, two different subset orbits can produce the same irreducible factor. This is exactly the mechanism in the characteristic-29 example, not a loss of separability of the original degree-seven equation.

Complementation gives a useful symmetry:

$$s_{A^c} = \text{Tr}(x_0) - s_A, \quad \text{Tr}(x_0) = \sum_{i=0}^{p-1} x_i.$$

Consequently separation for k is equivalent to separation for $p - k$. The cases $k = 1, p - 1$ always separate. The interesting cases are $2 \leq k \leq p - 2$.

3 An exact cyclotomic criterion for bad characteristics

For distinct $A, B \in \mathcal{S}_{p,k}$ define

$$q_{A,B}(T) = \sum_{i \in A} T^i - \sum_{i \in B} T^i, \quad \Phi_p(T) = 1 + T + \cdots + T^{p-1}, \quad (2)$$

and the positive integer

$$N_{A,B} = |\text{Res}_T(\Phi_p, q_{A,B})|. \quad (3)$$

Equivalently, $N_{A,B}$ is the absolute norm of $q_{A,B}(\zeta)$ from $\mathbb{Q}(\zeta)$, for a primitive p th root of unity ζ .

Lemma 3.1. *Every integer $N_{A,B}$ in (3) is nonzero. It is divisible by p .*

Proof. The polynomial $q = q_{A,B}$ is nonzero, has degree at most $p - 1$, and $q(1) = 0$. If q were divisible by Φ_p over $\mathbb{Q}$, it would equal $c\Phi_p$ for some constant c , and $0 = q(1) = cp$ would force $c = 0$. Irreducibility of Φ_p over $\mathbb{Q}$ therefore gives a nonzero resultant. Modulo p , $\Phi_p = (T - 1)^{p-1}$ and $q(1) = 0$, so the reductions have a common root. Their resultant is zero modulo p . $\square$

Definition 3.2. For $2 \leq k \leq p - 2$, let $\mathcal{B}_{p,k}$ be the set of prime divisors of the integers $N_{A,B}$, as $\{A, B\}$ runs over all unordered pairs of distinct k -subsets. It is a finite, effectively computable set.

Theorem 3.3 (Exact universal separation criterion). *Let p be an odd prime and $2 \leq k \leq p - 2$.*

- (i) *In characteristic zero, the k -subset sums of every separable irreducible degree- p polynomial separate.*
- (ii) *In characteristic $\ell \neq p$, the following are equivalent:*

- (a) for every field K of characteristic ℓ and every separable irreducible degree- p polynomial over K , the k -subset sums separate;
- (b) $\ell \notin \mathcal{B}_{p,k}$.

(iii) In characteristic p , universal separation fails.

Thus $\mathcal{B}_{p,k}$ is exactly the set of characteristics in which a counterexample can exist.

3.1 Proof of the safe direction

Let L/K be the splitting field of f , and $G = \text{Gal}(L/K)$. Irreducibility makes the root action transitive, so p divides $|G|$. Cauchy's theorem supplies an element of order p . A nonidentity order- p permutation on p letters is a p -cycle. We may consequently number the roots so that

$$\sigma(x_i) = x_{i+1}, \quad i \pmod{p}.$$

All pairs of subsets occur in the definition of $\mathcal{B}_{p,k}$, so changing the numbering does not change the criterion.

Suppose that $s_A = s_B$, and write $q = q_{A,B}$. Then

$$q(\sigma)x_0 = 0.$$

In characteristic zero, Lemma 3.1 gives $\gcd(q, \Phi_p) = 1$ in $\mathbb{Q}[T]$. In characteristic $\ell \notin \mathcal{B}_{p,k}$, the resultant is nonzero modulo ℓ , so the same coprimality holds in $\mathbb{F}_\ell[T]$. In either case, over the prime field there are polynomials a, b with

$$a(T)q(T) + b(T)\Phi_p(T) = 1.$$

Apply the corresponding operator identity to $(\sigma - 1)x_0$. Both terms on the left vanish: the first because $q(\sigma)x_0 = 0$, and the second because

$$\Phi_p(\sigma)(\sigma - 1) = \sigma^p - 1 = 0.$$

Hence $(\sigma - 1)x_0 = 0$, contradicting $x_1 \neq x_0$. This proves (i) and the implication (b) $\Rightarrow$ (a) of (ii).

Notice that no irreducibility assertion about Φ_p over K was used. In particular, the safe direction remains valid when K already contains all p th roots of unity.

3.2 Proof that every bad prime really occurs

Suppose $\ell \neq p$ divides $N_{A,B}$. The reductions of Φ_p and $q_{A,B}$ have a common root ζ in $\overline{\mathbb{F}}_\ell$. Since $\ell \neq p$, this is a primitive p th root of unity. Put

$$K = \mathbb{F}_\ell(\zeta)(t), \quad f(X) = X^p - t,$$

where t is transcendental. The polynomial is Eisenstein at t in $\mathbb{F}_\ell(\zeta)[t][X]$, hence irreducible over K . Its derivative is pX^{p-1} , so it is separable. If $\alpha^p = t$, its roots are $x_i = \alpha\zeta^i$, and

$$s_A - s_B = \alpha q_{A,B}(\zeta) = 0.$$

Moreover, $K(\alpha)/K$ is already its splitting field and has cyclic Galois group of order p . Thus the unsafe characteristic is witnessed even by a very simple solvable polynomial. This proves the converse in (ii).

For (iii), use the Artin–Schreier polynomial

$$f(X) = X^p - X - t \in \mathbb{F}_p(t)[X].$$

It is separable. To see its irreducibility, first note that $t \neq b^p - b$ for any $b \in \mathbb{F}_p(t)$: a pole of $b^p - b$ has order divisible by p , whereas t has a pole of order one at infinity. If α is a root, all its roots $\alpha + i$ lie in $\mathbb{F}_p(t)(\alpha)$; the Galois group embeds into the additive group $\mathbb{F}_p$ and therefore has order one or p . The absence of a root in the base field rules out order one.

For a k -subset A , its sum is

$$k\alpha + \sum_{i \in A} i.$$

There are at most p such values, but $\binom{p}{k} > p$ in the stated range of k . Some distinct subsets have equal sums. This finishes the proof of Theorem 3.3. $\square$

Remark 3.4 (The exact logical scope). A prime in $\mathcal{B}_{p,k}$ does not mean that every polynomial in that characteristic has a collision. It means that a collision is possible and that a characteristic-only universal guarantee is false. A particular resolvent can still be tested for separability by a gcd computation.

[write] **Related positive-characteristic material in the project.** Two formal developments of `Algebra/PolynomialFormulas` concern neighbouring phenomena at the prime 3 [11]; neither states a theorem of this section.

- **LazardAlternatingResolventCounterexample** (Lean and Rocq; claim crosswalk l. 202) refutes the assertion that the alternating resolvent $X^2 - \Delta$ of an irreducible polynomial is always separable: over $\mathbb{F}_3(t)$ the cubic $X^3 - t$ is irreducible but *inseparable* (derivative zero), its discriminant vanishes, and its alternating resolvent is X^2 . There the source polynomial itself is inseparable. The characteristic-29 phenomenon of Theorem 5.1 is different: the septic $X^7 - t$ is separable, and only the auxiliary resolvent acquires repeated factors.
- **LazardArtinSchreierRadicalCounterexample** (Lean and Rocq; crosswalk l. 200) uses the Artin–Schreier cubic $X^3 - X - t$, the case $p = 3$ of the witness $X^p - X - t$ in part (iii) above, to show that a cyclic, hence solvable, Galois group in characteristic 3 need not give solvability by radicals. It concerns radical towers, not subset-sum collisions.

3.3 A general relation-space interpretation

The same proof applies to a prescribed coefficient vector $c = (c_0, \dots, c_{p-1}) \in \mathbb{Z}^p$ with $\sum c_i = 0$ and $q_c(T) = \sum c_i T^i \neq 0$. Outside the prime divisors of $\text{Res}(\Phi_p, q_c)$, the relation $\sum c_i x_i = 0$ is impossible for a separable irreducible degree- p polynomial. Conversely, every non- p prime divisor gives a pure-binomial witness for that relation after adjoining the necessary roots of unity. Thus the subset-sum theorem is one instance of an exact integral relation-obstruction method, not merely a collection of examples.

For a fixed characteristic different from p , relations among the cyclically ordered roots form a module over $\mathbb{F}_\ell[T]/(T^p - 1)$. The cyclotomic component $\mathbb{F}_\ell[T]/(\Phi_p)$ detects the nonconstant part. Our resultant test asks whether a prescribed small coefficient vector can vanish on this component. This viewpoint suggests extensions to cyclic codes and more general weighted resolvents, discussed in Section 14.

4 Complete classifications in degrees five and seven

Theorem 4.1 (Small-degree exceptional sets). *The exact exceptional characteristics are*

$$\boxed{\mathcal{B}_{5,2} = \{5\}}, \quad \boxed{\mathcal{B}_{7,2} = \{2, 7\}}, \quad \boxed{\mathcal{B}_{7,3} = \{2, 7, 29\}}.$$

The complementary subset sizes have the same exceptional sets.

We give the finite certificate behind this theorem explicitly. It is useful both for independent review and for a future formalization.

4.1 Reduction by the affine group

The affine group $\text{AGL}_1(\mathbb{F}_p)$ acts on subset labels by $i \mapsto ai + b$, where $a \neq 0$. For a primitive p th root ζ ,

$$q_{aA+b, aB+b}(\zeta) = \zeta^b q_{A,B}(\zeta^a).$$

The cyclotomic field automorphism $\zeta \mapsto \zeta^a$ preserves absolute norms, and ζ^b has norm one. Interchanging A and B changes only a sign. Therefore $N_{A,B}$ is constant on affine orbits of unordered pairs.

For each pair, choose the lexicographically smallest representative of its affine orbit. In the following tables, 012 denotes $\{0, 1, 2\}$, and similarly for other strings. The orbit sizes add to the number of all unordered pairs; disjointness is checked by the displayed normalization rule. Each norm can be computed as an integer determinant without algebraic number arithmetic.

4.2 The determinant certificate

Write

$$q(T)T^j \equiv \sum_{i=0}^{p-2} m_{ij}T^i \pmod{\Phi_p(T)}, \quad 0 \leq j \leq p-2.$$

Multiplication by q in $\mathbb{Z}[T]/(\Phi_p)$ has matrix $M_q = (m_{ij})$. Its determinant is $\text{Res}(\Phi_p, q)$. Reduction uses only

$$T^p = 1, \quad T^{p-1} = -(1 + T + \cdots + T^{p-2}).$$

These identities supply a particularly small exact certificate.

For the exceptional septic pair $A = 012$, $B = 346$, the matrix is

$$M_q = \begin{pmatrix} 2 & -1 & 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 1 & -2 & -2 \\ 2 & 1 & 2 & 0 & -1 & -2 \\ 0 & 1 & 2 & 2 & -2 & -1 \\ 0 & -1 & 2 & 2 & 0 & -2 \\ 1 & -1 & 0 & 2 & 0 & 0 \end{pmatrix}, \quad |\det M_q| = 203 = 7 \cdot 29. \quad (4)$$

The integer matrices for every other row are included in `certificates/cyclotomic_norm_certificates.json`.¹

A	B	Orbit size	$N_{A,B}$
01	02	20	5
01	04	10	5
01	23	10	5
01	24	5	25
Total, quintic pairs		45	

A	B	Orbit size	$N_{A,B}$
01	02	42	7
01	03	42	7
01	06	21	7
01	23	21	7
01	24	42	56
01	26	21	49
01	34	21	7
Total, septic pairs		210	

¹[write] Shipped in this report as `data/cyclotomic_norm_certificates.json`.

Table 1: All affine orbits of unordered pairs of septic triples.

Row	A	B	Orbit size	$N_{A,B}$
1	012	013	42	7
2	012	014	42	7
3	012	015	42	7
4	012	016	21	7
5	012	023	42	7
6	012	034	42	7
7	012	035	42	56
8	012	045	42	7
9	012	046	42	49
10	012	056	21	7
11	012	134	42	56
12	012	136	21	49
13	012	345	21	7
14	012	346	42	203
15	013	015	21	7
16	013	026	42	56
17	013	046	21	7
18	013	245	7	343
Total			595	

Proof of Theorem 4.1. The tables exhaust the pair sets of sizes $\binom{10}{2} = 45$, $\binom{21}{2} = 210$, and $\binom{35}{2} = 595$. The norms occurring in the three cases are, respectively,

$$\{5, 25\}, \quad \{7, 49, 56\}, \quad \{7, 49, 56, 203, 343\}.$$

Their prime divisors are exactly the displayed exceptional sets. Theorem 3.3 supplies both the universal safe conclusion and the existence of counterexamples in every exceptional characteristic. $\square$

As an additional checksum on the finite enumeration, the complete norm multiplicities are

$$\begin{aligned} (5, 2) : & \quad 5^{[40]}, 25^{[5]}, \\ (7, 2) : & \quad 7^{[147]}, 49^{[21]}, 56^{[42]}, \\ (7, 3) : & \quad 7^{[357]}, 49^{[63]}, 56^{[126]}, 203^{[42]}, 343^{[7]}. \end{aligned}$$

Here $n^{[m]}$ means that norm n occurs m times, not an exponentiation. The independent auditor regenerates the orbits and multiplication matrices, checks every determinant, and verifies that their union is exactly the full pair set. No inference from a sample of pairs is being made.

5 Explicit exceptional-characteristic witnesses

5.1 The characteristic-29 septic

Let $K = \mathbb{F}_{29}(t)$, where t is transcendental, and let $\alpha^7 = t$. Take $\zeta = 23 \in \mathbb{F}_{29}$. Its successive powers are

$$(1, 23, 7, 16, 20, 25, 24), \quad \zeta^7 = 1.$$

They are distinct, so ζ is primitive. The polynomial

$$f(X) = X^7 - t$$

is irreducible by Eisenstein, is separable because its derivative is $7X^6$, and has splitting field $K(\alpha)$ with cyclic group C_7 . Its roots are $x_i = \alpha\zeta^i$.

The following equality is entirely explicit:

$$1 + 23 + 7 \equiv 16 + 20 + 24 \equiv 2 \pmod{29}.$$

Thus

$$x_0 + x_1 + x_2 = x_3 + x_4 + x_6. \quad (5)$$

The roots in the two triples are all different. This is not a collision caused by an overlap or by a repeated root of f .

Theorem 5.1 (Exact repeated-factor resolvent). *Over $\mathbb{F}_{29}(t)$,*

$$\begin{aligned} R_{7,3,X^7-t}(Y) &= (Y^7 - 12t)^2(Y^7 - t)(Y^7 - 17t)(Y^7 - 28t) \\ &= (Y^7 - 12t)(Y^{28} - t^4). \end{aligned} \quad (6)$$

Every displayed degree-seven binomial is irreducible. The orbit product has degree 35, but its squarefree part has degree 28.

Proof. There are five translation orbits on the 35 triples. For a representative A , set

$$c_r(A) = e_r(\zeta^i : i \in A), \quad r = 1, 2, 3.$$

The representative values and their seventh powers are as follows; the second and third coordinates will also be used for the repair below.

A	c_1	c_2	c_3	c_1^7
012	2	17	16	12
013	11	1	20	12
014	15	10	25	17
015	20	14	24	1
024	28	22	24	28

Translation by b multiplies the sum $c_1\alpha$ by ζ^b . The product for one orbit is therefore $Y^7 - c_1^7t$. Multiplying the five orbit products gives the first line of (6). The four constants 1, 12, 17, 28 are the fourth roots of unity in $\mathbb{F}_{29}$, which gives the second line. Each nonzero binomial $Y^7 - ct$ is Eisenstein at t , hence irreducible. The four constants are distinct, so the asserted squarefree degree follows. $\square$

5.2 Characteristic two: the Fano-plane degeneration

Let $K = \mathbb{F}_8(t)$, let ζ generate $\mathbb{F}_8^\times$, and again let $\alpha^7 = t$. The polynomial $X^7 - t$ is separable and irreducible over K . Its seven nonzero root coefficients are all of $\mathbb{F}_8^\times$.

A triple of distinct nonzero field elements has sum zero precisely when it is $\{a, b, a+b\}$. There are $7 \cdot 6/6 = 7$ such triples, the lines of the Fano plane. Multiplication by $\mathbb{F}_8^\times$ permutes the other triples and acts transitively on their nonzero sums. The remaining 28 triples are therefore distributed equally, four per nonzero sum. Hence

$$R_{7,3,X^7-t}(Y) = Y^7(Y^7 - t)^4 \quad \text{over } \mathbb{F}_8(t). \quad (7)$$

The auxiliary product has only eight distinct roots, even though the original polynomial has seven distinct roots and a cyclic Galois group.

For septic pair sums in characteristic two, the same field gives $\zeta^a + \zeta^b = \zeta^c + \zeta^d$ for suitable distinct pairs, in agreement with $2 \in \mathcal{B}_{7,2}$. For instance, in $\mathbb{F}_2[z]/(z^3 + z + 1)$, the pairs of field elements $\{1, z\}$ and $\{z^2, 1 + z + z^2\}$ have equal sum.

5.3 Characteristic seven: an Artin–Schreier degeneration

Over $\mathbb{F}_7(t)$, let $f = X^7 - X - t$ and choose a root α . The irreducibility argument in Section 3 applies, and the roots are $\alpha + i$ for $i \in \mathbb{F}_7$. A triple sum is $3\alpha + c$. Translation of a triple by b changes c by $3b$, so each of the seven values occurs exactly five times. Moreover,

$$(3\alpha)^7 - 3\alpha = 3t.$$

It follows that

$$R_{7,3,X^7-X-t}(Y) = (Y^7 - Y - 3t)^5. \quad (8)$$

Likewise,

$$R_{5,2,X^5-X-t}(Y) = (Y^5 - Y - 2t)^2 \quad \text{over } \mathbb{F}_5(t). \quad (9)$$

These identities make the exceptional prime-degree characteristic concrete, rather than merely noting that Fourier inversion cannot divide by p .

6 The exact implication for published septic factor patterns

Proposition 7 and the subsequent specialized discussion in Faggal–Lazard describe a cyclic septic case using five degree-seven factors; Section 5.2 permits characteristic 29 under its stated exclusions [2]. Equation (6) shows that one cannot universally interpret those factors as five *distinct* irreducible factors attached to five distinguishable subset orbits. Two orbit products coincide.

If factors are counted with multiplicity, our example still has five irreducible factors in that weaker sense. The failure is therefore stated precisely: the unqualified separation needed to identify different subset orbits from different scalar factors is false. Also, the degree-35 orbit product is not a minimal polynomial of any one triple sum; in this example each individual triple sum has degree seven.

The sharp repair for this particular separation step is now known:

$$\text{char } K \notin \{2, 7, 29\}.$$

This assertion follows for *every* irreducible separable septic from Theorem 4.1, without a solvability assumption. When retaining the separate characteristic restrictions of a radical construction, one must take their union with these restrictions. Conversely, an input-specific squarefreeness check is sufficient for the orbit-to-factor principle even in a bad characteristic.

This is not a complete audit of all matching predicates or denominator conditions used in the published algorithm. Additional root relations can have different cyclotomic norm obstructions. Excluding 29 repairs the triple-sum separation claim; it does not by itself certify every subsequent formula. The next section offers a repair that avoids relying on a characteristic-only exclusion at all.

7 A uniform separating-invariant repair

Let $x_1, \dots, x_n$ be pairwise distinct elements of a field extension of K . Fix $1 \leq k < n$, and set $N = \binom{n}{k}$. For each k -subset A let

$$E(A) = (e_1(x_A), \dots, e_k(x_A)),$$

where e_r is the r th elementary symmetric function of the elements whose indices lie in A .

Lemma 7.1 (Subset descriptors). *The map $A \mapsto E(A)$ is injective in every characteristic.*

Proof. Its coordinates determine the monic polynomial

$$\prod_{i \in A} (Z - x_i) = Z^k - e_1(x_A)Z^{k-1} + \dots + (-1)^k e_k(x_A).$$

Equal coordinate vectors give equal polynomials and therefore equal sets of roots. Pairwise distinctness of the original x_i then gives equal index subsets. $\square$

Introduce a parameter u and compress this vector to the scalar

$$h_A(u) = \sum_{r=1}^k u^{r-1} e_r(x_A). \quad (10)$$

For different subsets A, B , the difference $h_A - h_B$ is a nonzero polynomial in u of degree at most $k - 1$.

Theorem 7.2 (Finite separating family). *At most*

$$B_{n,k} = (k-1) \binom{N}{2}, \quad N = \binom{n}{k}, \quad (11)$$

values of u in an algebraic closure of the field containing the x_i can cause two of the $h_A(u)$ to coincide. Consequently, any $B_{n,k} + 1$ distinct values in K include one for which all the $h_A(u)$ are distinct.

If the x_i are the roots of a separable polynomial over K , then

$$H_u(Y) = \prod_{|A|=k} (Y - h_A(u)) \in K[Y] \quad (12)$$

for every $u \in K$. At a separating parameter, its irreducible factors recover the exact Galois orbit sizes on k -subsets.

Proof. Each of the $\binom{N}{2}$ nonzero difference polynomials has at most $k - 1$ roots. Taking their union gives (11), with no restriction on characteristic. The product (12) is invariant under every permutation of the x_i , so its coefficients are symmetric polynomials in the roots and hence belong to K . At a separating parameter, Lemma 2.2 applies. $\square$

[write] Existing formal counterparts. The project already formalizes this strategy for sextic block systems in `Algebra/PolynomialFormulas/Lean/PolynomialFormulas/SexticSeparatingInvariants.lean` [9], whose docstring starts from the same observation that scalar pair and triple invariants can collide after specialization. `rootPolynomial_injective` (l. 115) is the argument of Lemma 7.1 for six roots and arbitrary index subsets, over any commutative domain (no characteristic hypothesis). `exists_nat_eval_injective` (l. 1048) separates a finite injective family of multivariate polynomials by one natural-number evaluation point, via the combinatorial Nullstellensatz, over a *characteristic-zero* domain, and `exists_pairDescriptor_separating_evaluation` and `exists_tripleDescriptor_se`

`parating_evaluation` (l. 1074, 1083) apply it to the sextic block descriptors. The resulting “proved terminating unbounded search” is described in `Algebra/PolynomialFormulas/Decidability.md` (l. 284–290). These are analogues of the declarations `elementarySubsetDescriptor_injective` and `separatingParameter_exists` proposed in Section 13. What Lemma 7.1 and Theorem 7.2 add is the general (n, k) setting, the one-parameter compression $h_A(u)$, and an explicit hitting bound $B_{n,k}$ valid in every characteristic.

For septics and triples,

$$N = 35, \quad B_{7,3} = 2 \binom{35}{2} = 1190.$$

Thus 1191 *distinct* parameters always suffice whenever the base field has at least that many elements. Over $\mathbb{Q}$ one may use $0, 1, \dots, 1190$. In characteristic 29, those same integer numerals would repeat and would not constitute a valid hitting set. Over $\mathbb{F}_{29}(t)$ one can instead use $1, t, t^2, \dots, t^{1190}$.

Over a finite field with fewer elements, the theorem does not assert the existence of a separating parameter in that field. Passing to a sufficiently large finite extension, or keeping u transcendental, supplies distinct parameters; the associated change of base field must then be recorded when interpreting Galois factor patterns.

7.1 How to compute the repaired product without knowing the roots

Regard $x_1, \dots, x_n$ first as indeterminates. The product (12) is a symmetric polynomial in them, with coefficients in $\mathbb{Z}[u, Y]$. The fundamental theorem of symmetric polynomials expresses it as a polynomial in the elementary symmetric functions of the full root set. Substitution of the coefficients of the original monic polynomial therefore computes H_u over its coefficient field. This is an algebraic construction; it does not require numerical roots or a pre-existing radical solution.

A deterministic specialization procedure is consequently available: compute H_u for each chosen parameter, test $\gcd(H_u, H'_u) = 1$, and stop at the first success. Theorem 7.2 proves termination within the stated number of distinct candidates. It is a correctness and termination bound, not a claim of good asymptotic complexity. Expanding the universal symmetric polynomial can be expensive.

7.2 Power-sum variants and their characteristic restriction

When $1, \dots, k$ are invertible in K , Newton identities show that the vector

$$\left(\sum_{i \in A} x_i, \sum_{i \in A} x_i^2, \dots, \sum_{i \in A} x_i^k \right)$$

also distinguishes subsets. Then one may use

$$\sum_{i \in A} (x_i + ux_i^2 + \dots + u^{k-1}x_i^k).$$

This has the appealing form of an ordinary subset sum after a common polynomial transformation of the roots. The elementary-symmetric version (10), however, avoids division by $2, \dots, k$ and works in every characteristic. These two versions should not be silently identified in small characteristic.

7.3 Repairing the characteristic-29 example explicitly

For $X^7 - t$ over $\mathbb{F}_{29}(t)$, use $u = 1$. Then

$$h_A(1) = c_1(A)\alpha + c_2(A)\alpha^2 + c_3(A)\alpha^3.$$

The 35 triples have distinct coefficient vectors (c_1, c_2, c_3) by Lemma 7.1, applied to the seven distinct powers of ζ . Since $1, \alpha, \dots, \alpha^6$ are linearly independent over $\mathbb{F}_{29}(t)$, all 35 displayed values are distinct. Thus one nonzero parameter already repairs this particular example.

The five now-distinct irreducible factors have the following compact exact specification. For each of the five rows (c_1, c_2, c_3) in the table in Theorem 5.1, take

$$Q_A(Y) = \text{Res}_Z(Z^7 - t, Y - c_1Z - c_2Z^2 - c_3Z^3). \quad (13)$$

Each Q_A is monic of degree seven, and $H_1 = \prod_A Q_A$. They are distinct and irreducible by Lemma 2.2. This is a coefficient-level description of the repaired factors, not merely a statement that some repair exists.

For completeness, the affine solvable septic groups give the following orbit patterns, now valid for any separating equivariant descriptor:

Group	Factor degrees
C_7	7, 7, 7, 7, 7
$C_7 \rtimes C_2$	7, 7, 7, 14
$C_7 \rtimes C_3$	7, 7, 21
$C_7 \rtimes C_6$	14, 21

These counts follow directly by acting on the 35 triples with $i \mapsto ai + b$, where the allowed multipliers form the indicated subgroup of $\mathbb{F}_7^\times$. The verifier also checks these four finite actions.

8 One-radical Fourier–Kummer reconstruction

We now address a different failure mode: reconstructing roots from modes without incompatible radical branches. Assume throughout this section that $\text{char } E \neq p$, that $\zeta \in E$ is a primitive p th root of unity, and that $\sigma \in \text{Aut}(E)$ has order p and fixes ζ . Let $B = E^{\langle \sigma \rangle}$ and suppose that

$$\sigma(x_i) = x_{i+1}, \quad i \pmod{p}.$$

We use the Fourier convention

$$u_j = \sum_{i=0}^{p-1} \zeta^{-ij} x_i, \quad x_i = \frac{1}{p} \sum_{j=0}^{p-1} \zeta^{ij} u_j. \quad (14)$$

This differs by an index sign from the positive-exponent forward convention in the inspected Lean module. Substituting its character by its inverse aligns the conventions exactly; no mathematical change is involved.

The finite geometric-sum identity gives

$$\sigma(u_j) = \zeta^j u_j, \quad u_0 \in B.$$

If the x_i are not all equal, at least one u_j with $j \neq 0$ is nonzero. This is the only existence condition needed for a pivot chart.

For $j \in \{1, \dots, p-1\}$ define $e_j(k)$ to be the unique integer in $\{1, \dots, p-1\}$ satisfying

$$je_j(k) \equiv k \pmod{p}, \quad 1 \leq k \leq p-1.$$

Theorem 8.1 (Coherent one-radical chart). *Assume $u_j \neq 0$. Put*

$$a_j = u_j^p, \quad c_{j,k} = \frac{u_k}{u_j^{e_j(k)}} \quad (1 \leq k \leq p-1). \quad (15)$$

Then $a_j \in B^\times$, $c_{j,k} \in B$, and $c_{j,j} = 1$. For any root ρ of $T^p - a_j$ in an algebraic closure of E , define

$$y_i = \frac{1}{p} \left(u_0 + \sum_{k=1}^{p-1} \zeta^{ik} c_{j,k} \rho^{e_j(k)} \right). \quad (16)$$

There exists $r \in \mathbb{F}_p$ such that

$$y_i = x_{i+rj-1} \quad \text{for every } i \in \mathbb{F}_p. \quad (17)$$

In particular, all branches give the same unordered root set, with no additional independent p th-root choices.

Proof. The covariance formula gives

$$\sigma(a_j) = a_j, \quad \sigma(c_{j,k}) = \zeta^{k-j e_j(k)} c_{j,k} = c_{j,k}.$$

The nonzero denominator is part of the hypothesis, not a convention about division by zero. Since E already contains u_j and all p th roots of unity, every root of $T^p - a_j$ is $\rho = \zeta^r u_j$ for some r . Substitution in (16) yields

$$c_{j,k} \rho^{e_j(k)} = \zeta^{r e_j(k)} u_k = \zeta^{r j^{-1} k} u_k.$$

Fourier inversion now gives (17) simultaneously for all i . Thus coherence is derived, not assumed separately for each mode. $\square$

Corollary 8.2 (Cyclic extension generated by a pivot). *Under the hypotheses of Theorem 8.1, $[B(u_j) : B] = p$. If E/B is the cyclic extension of degree p , then $E = B(u_j)$ and $T^p - a_j$ is the minimal polynomial of u_j over B .*

Proof. The p values $\sigma^r(u_j) = \zeta^{rj} u_j$ are distinct. Thus its minimal polynomial over B has at least p roots, while $T^p - a_j$ gives an upper bound of p on its degree. The remaining assertion follows by the degree formula. $\square$

This is a relative minimality statement: the cyclic degree- p step really is nontrivial, and one p th radical suffices over B . It is *not* a claim that an entire radical expression over the original coefficient field uses the smallest possible number of radicals. The repository's warnings about stronger optimality assertions remain relevant.

[write] **An abstract counterpart in Rocq.** Algebra/PolynomialFormulas/Coq/LazardCyclicKummerGenerator.v proves `cyclic_kummer_generator` (l. 32): if F/E is cyclic Galois of degree n and E contains a primitive n th root of unity, there is a nonzero $x \in F$ with $x^n \in E$ and $F = E(x)$ [10]. It is the abstract existence form of Corollary 8.2; the corollary adds that any nonzero Fourier mode u_j is such a generator.

8.1 Transition maps on overlaps

Suppose $u_j, u_h \neq 0$, and set $e = e_j(h)$. Then

$$u_h = c_{j,h} u_j^e, \quad a_h = c_{j,h}^p a_j^e.$$

A branch $\rho_j^p = a_j$ determines the compatible branch

$$\rho_h = c_{j,h}\rho_j^e, \quad \rho_h^p = a_h. \quad (18)$$

It gives the same reconstructed ordered tuple when the chart coordinate conventions are matched. More explicitly, if $\rho_j = \zeta^r u_j$, then $\rho_h = \zeta^{re} u_h$, and the shift in the h -chart is $re h^{-1} = rj^{-1}$ modulo p .

For the invariant coefficients, let $f = e_h(k)$ and $g = e_j(k)$. The integer

$$m = \frac{ef - g}{p}$$

is well defined. Direct substitution gives the exact transition rule

$$c_{h,k} = c_{j,k} c_{j,h}^{-f} a_j^{-m}. \quad (19)$$

This exhibits an actual atlas, with compatible local coordinates, rather than unrelated formulas whose radical branches must be guessed anew.

8.2 What the chart theorem does not compute

The input to Theorem 8.1 includes the invariant data $u_0, a_j, c_{j,k}$ over B . Constructing these efficiently from the original polynomial coefficients is a separate invariant-theoretic task. The theorem does not hide that task inside a claim of an executable coefficient-only solver. Its immediate use in ProveIt is as a small correctness theorem between an invariant-recovery stage and an already formalized Fourier reconstruction stage.

9 Solvable prime-degree groups and the coefficient-field tower

We record the precise field-theoretic setting in which the cyclic chart arises from a solvable prime-degree equation. The group-theoretic argument is included to make the dependence explicit.

Lemma 9.1 (Solvable transitive groups of prime degree). *A solvable transitive subgroup $G \leq S_p$ is conjugate to $C_p \rtimes C_d \leq \text{AGL}_1(\mathbb{F}_p)$ for some $d \mid p-1$.*

Proof. A transitive action of prime degree is primitive: every block size divides p . Choose a minimal nontrivial normal subgroup N of G . Solvability implies that N is elementary abelian: its commutator subgroup is characteristic in N and normal in G , so minimality and solvability make it trivial. The primary subgroups of the resulting finite abelian group are characteristic, forcing a single prime; multiplication by that prime has characteristic image and cannot be surjective, so this image is zero. Its orbits form a block system, so N is transitive; otherwise every orbit would be a singleton, contrary to faithfulness and $N \neq 1$.

An abelian transitive permutation group is regular. Indeed, an element fixing one point fixes every point by commuting with a transitive group. Thus $|N| = p$ and $N \cong C_p$. Numbering the points by $\mathbb{F}_p$ identifies N with the translations. Its normalizer in S_p is the affine group: a normalizing permutation is determined by the image of 0 and the nonzero multiplier by which it conjugates a generator of translations. Hence G lies in that normalizer. Its multiplier image is a subgroup of $\mathbb{F}_p^\times$, therefore cyclic of order $d \mid p-1$. Since all translations are present, subtracting the translation part of each affine element gives the asserted semidirect product. $\square$

Now let $f \in K[X]$ be separable irreducible of degree p in characteristic zero, with solvable Galois group $G = C_p \rtimes C_d$, and let L be its splitting field. Write

$$B_0 = L^{C_p}, \quad E = L(\zeta), \quad B = B_0(\zeta).$$

Then B_0/K is cyclic of degree d , and B/K is an abelian Galois extension, as the compositum of B_0/K with a cyclotomic extension. Moreover,

$$[B : K] \leq d(p-1) \leq (p-1)^2.$$

Because $[B : B_0]$ divides $p-1$, it is relatively prime to $[L : B_0] = p$. Thus $L \cap B = B_0$ and

$$[E : B] = p.$$

The cyclic generator on L/B_0 extends to E fixing B , so all the hypotheses of the Fourier–Kummer chart theorem are available after numbering the roots by this generator.

Corollary 9.2 (Field-theoretic solver decomposition). *In characteristic zero, a solvable irreducible prime-degree equation admits the following decomposition: construct the abelian field B of degree at most $(p-1)^2$ over K , select a nonzero Fourier mode, and perform the single cyclic p th-radical reconstruction of Theorem 8.1. For $p = 5$ the displayed bound is 16; for $p = 7$ it is 36.*

These are field-degree bounds, not optimal radical-count bounds. The abelian stage itself is solvable by radicals by the classical Galois construction after adjoining suitable roots of unity. A fast implementation must exploit the smaller actual Galois group and its invariants rather than blindly realize the worst-case degree bound.

In positive characteristic, the same cyclic p -chart remains valid when $\text{char } K \neq p$ and the stated cyclic extension and root-of-unity data are supplied. The equivalence between solvable groups and ordinary radical towers needs its own characteristic hypotheses and is not inferred merely from group solvability here.

10 Every nonempty Fourier support really occurs

A formula that works for generic tuples need not survive vanishing Fourier modes. The following construction shows that no nonempty pattern can be excluded solely from rational irreducibility or from maximal solvable Galois group.

Theorem 10.1 (Full support realization). *Let p be an odd prime, let $\emptyset \neq S \subseteq \{1, \dots, p-1\}$, and choose nonzero rational numbers c_k for $k \in S$. Let $\alpha = 2^{1/p}$ and let ζ be a primitive p th root of unity. Put*

$$P(Z) = \sum_{k \in S} c_k Z^k, \quad x_i = P(\alpha \zeta^i), \quad 0 \leq i \leq p-1. \quad (20)$$

The x_i are the roots of an irreducible rational polynomial of degree p . Its splitting field has Galois group $\text{AGL}_1(\mathbb{F}_p)$. Relative to the indicated cyclic numbering, its Fourier modes are

$$u_0 = 0, \quad u_k = \begin{cases} pc_k \alpha^k, & k \in S, \\ 0, & k \notin S. \end{cases} \quad (21)$$

Thus all $2^{p-1} - 1$ nonempty supports occur.

Proof. The polynomial $Z^p - 2$ is Eisenstein at 2, so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = p$. Linear independence of $1, \alpha, \dots, \alpha^{p-1}$ shows that $P(\alpha) \notin \mathbb{Q}$. Since p is prime, the tower formula forces

$$\mathbb{Q}(P(\alpha)) = \mathbb{Q}(\alpha).$$

Therefore $P(\alpha)$ has degree p , and its images under the p embeddings of $\mathbb{Q}(\alpha)$ are exactly the displayed distinct x_i . The normal closures of $\mathbb{Q}(P(\alpha))$ and $\mathbb{Q}(\alpha)$ coincide.

For completeness, $[\mathbb{Q}(\zeta) : \mathbb{Q}] = p - 1$ and $\alpha \notin \mathbb{Q}(\zeta)$: otherwise the degree p would divide $p - 1$. Over $\mathbb{Q}(\zeta)$, adjoining α gives the splitting field of $Z^p - 2$; its Galois group embeds in μ_p , and nontriviality forces degree p . Hence the full normal closure has degree $p(p - 1)$ and its natural affine permutation action is all of $\text{AGL}_1(\mathbb{F}_p)$.

Finally, substitute (20) into (14). The geometric sum $\sum_i \zeta^{i(k-j)}$ equals p when $k = j$ and zero otherwise, giving (21). $\square$

Corollary 10.2 (Sharpness of the coordinate-pivot cover). *For a fixed primitive root of unity and a fixed cyclic numbering, none of the $p - 1$ charts $u_j \neq 0$ can be omitted if the chart family is required to cover every irreducible rational degree- p example with full affine Galois group.*

Proof. Use Theorem 10.1 with $S = \{j\}$. Only the j th chart is available. $\square$

The qualifiers matter. This is a minimality statement for this specific coordinate-pivot cover, not a lower bound for every conceivable nonlinear cover. Relabelling the cyclic generator permutes the nonzero Fourier indices; choosing such a relabelling adaptively is another form of chart selection, not a contradiction.

10.1 Concrete rational examples

Taking every $c_k = 1$, elimination of Z from $Z^p - 2 = 0$ and $Y - P(Z) = 0$ gives the following monic polynomials:

$$\begin{aligned} p = 5, S = \{1, 2\} : & Y^5 - 10Y^2 - 10Y - 6, \\ p = 5, S = \{2, 3\} : & Y^5 - 10Y^3 + 20Y - 12, \\ p = 5, S = \{1, 2, 3, 4\} : & Y^5 - 20Y^3 - 60Y^2 - 70Y - 30, \\ p = 7, S = \{1, 2\} : & Y^7 - 14Y^3 - 28Y^2 - 14Y - 6, \\ p = 7, S = \{2, 3\} : & Y^7 - 14Y^4 - 28Y^2 + 28Y - 12, \\ p = 7, S = \{1, \dots, 6\} : & Y^7 - 42Y^5 - 210Y^4 - 490Y^3 \\ & - 630Y^2 - 434Y - 126. \end{aligned}$$

Their irreducibility and Galois groups follow from Theorem 10.1, not from numerical root matching. The exact resultants are also regenerated by the verifier.

For example, in the first septic case, $u_1 = 7\alpha$ and $u_2 = 7\alpha^2$, while all other nonconstant modes vanish. In the $j = 1$ chart,

$$a_1 = 7^7 \cdot 2, \quad c_{1,1} = 1, \quad c_{1,2} = \frac{1}{7}.$$

The formula becomes

$$y_i = \frac{1}{7} \left(\zeta^i \rho + \frac{\zeta^{2i}}{7} \rho^2 \right), \quad \rho^7 = 7^7 \cdot 2.$$

Choosing $\rho = 7\alpha\zeta^r$ recovers $x_{i+r} = \alpha\zeta^{i+r} + \alpha^2\zeta^{2(i+r)}$ exactly.

11 Rigidity of the exceptional septic configurations

The bad-prime calculation can say more than where counterexamples exist. In characteristics 29 and 2, the small number of surviving cyclotomic characters forces the entire root configuration into a special form.

11.1 Characteristic 29: precisely the translated binomial locus

Theorem 11.1 (Complete characteristic-29 collision classification). *Let K be any field of characteristic 29, and let $f \in K[X]$ be monic, separable, and irreducible of degree seven. The following are equivalent:*

- (i) *Two different triples of roots of f have the same sum.*
- (ii) *For some $\eta, b \in K$ with $b \neq 0$,*

$$f(X) = (X - \eta)^7 - b.$$

Whenever these conditions hold, the Galois group of f is C_7 , and

$$R_{7,3,f}(Y) = ((Y - 3\eta)^7 - 12b)((Y - 3\eta)^{28} - b^4). \quad (22)$$

In particular, every noncyclic irreducible septic over a field of characteristic 29 has separating triple sums.

Proof. The primitive seventh root $\zeta = 23$ lies in the prime field, hence in K . Choose a Galois seven-cycle σ and number the roots by it. Suppose a collision gives $q(\sigma)x_0 = 0$, with $q = q_{A,B}$ as in (2). The nonzero-character part of q must have a zero: otherwise the Bézout argument in Theorem 3.3 would rule out the collision.

Table 1 shows that a pair with a resultant divisible by 29 is affine-equivalent to $A = 012$, $B = 346$. For the representative

$$q_*(T) = 1 + T + T^2 - T^3 - T^4 - T^6,$$

its values at the six nontrivial seventh roots are

$$(q_*(\zeta), q_*(\zeta^2), \dots, q_*(\zeta^6)) = (0, 14, 18, 25, 17, 20) \quad \text{in } \mathbb{F}_{29}.$$

There is exactly one zero. An affine change of the subset labels multiplies values by a nonzero root of unity and permutes the nontrivial characters, so every potentially colliding pair has exactly one such zero. Call its index j .

Project the relation $q(\sigma)x_0 = 0$ to the r th character by applying $\sum_{i=0}^6 \zeta^{-ir} \sigma^i$. Since σ fixes ζ , the result is

$$q(\zeta^r)u_r = 0.$$

Thus every nonconstant Fourier mode except possibly u_j vanishes. At least one is nonzero, so $u_j \neq 0$. Fourier inversion gives

$$x_i = \eta + \beta \zeta^{ij}, \quad \eta = \frac{u_0}{7}, \quad \beta = \frac{u_j}{7}.$$

The trace shows $\eta \in K$. The full root product is $(X - \eta)^7 - \beta^7$, and its coefficients show that $b = \beta^7 \in K$. This proves (i) $\Rightarrow$ (ii).

Conversely, all roots of a polynomial in (ii) are $\eta + \beta \zeta^i$. The characteristic-29 equality (5) persists after adding η to each root, since both sides contain three roots. Hence there is a triple-sum collision. The original polynomial is irreducible, and K contains μ_7 , so $K(\beta)$ is its cyclic degree-seven splitting field. Finally, translating each root adds 3η to each triple sum. The five-orbit calculation in Theorem 5.1, with t replaced by b , gives (22). $\square$

The conclusion is stronger than adding 29 to a blanket list of excluded characteristics. An exact solver in that characteristic can first test whether its input becomes a pure binomial after removing the trace. If it does, the equation has an immediate Kummer solution and the repeated-factor resolvent is understood completely. If it does not, the ordinary triple-sum resolvent separates. Cyclicity alone is not sufficient for a collision; the specific primitive element must give the translated binomial form.

11.2 Characteristic two: affine Fano configurations

A polynomial $L(Z) = \sum_i a_i Z^{2^i}$ in characteristic two is *additive*: $L(v + w) = L(v) + L(w)$. Its roots, when distinct, form a vector space over $\mathbb{F}_2$. This standard structure is exactly the one forced by a septic subset-sum collision.

Theorem 11.2 (Complete characteristic-two collision classification). *Let K have characteristic two, and let $f \in K[X]$ be monic, separable, and irreducible of degree seven. The following are equivalent:*

- (i) *Two different pairs of roots have equal sums.*
- (ii) *Two different triples of roots have equal sums.*
- (iii) *For some $\eta, a, b, d \in K$, with $d \neq 0$,*

$$f(X) = F(X - \eta), \quad F(Z) = Z^7 + aZ^3 + bZ + d. \quad (23)$$

In this case the roots are $\eta + (W \setminus \{0\})$, where W is a three-dimensional $\mathbb{F}_2$ -subspace of the splitting field. The center η is the trace of a root, and

$$R_{7,2,f}(Y) = F(Y)^3, \quad R_{7,3,f}(Y) = (Y - \eta)^7 f(Y)^4. \quad (24)$$

Its Galois group embeds in $\text{GL}_3(\mathbb{F}_2)$.

Proof. Adjoin a primitive seventh root ζ . This extends the base field by degree one or three, relatively prime to seven, so f remains irreducible: for a root field M/K of degree seven, the degree of $MK(\zeta)/K$ is divisible by both 7 and $[K(\zeta) : K]$ and is at most their product, forcing equality. Its splitting-field group over $K(\zeta)$ consequently contains a seven-cycle σ fixing ζ .

Assume (i) or (ii). The collision yields a nonzero polynomial $q \in \mathbb{F}_2[T]$ with $\deg q \leq 6$, $q(1) = 0$, and $q(\sigma)x_0 = 0$. In characteristic two,

$$\Phi_7(T) = (T^3 + T + 1)(T^3 + T^2 + 1),$$

with both cubic factors irreducible. The gcd of q and Φ_7 cannot be one, by the safe-direction proof of Theorem 3.3. It cannot be all of Φ_7 either: degree at most six would give $q = \Phi_7$, contradicting $q(1) = 0$ and $\Phi_7(1) = 1$. Therefore it is exactly one of the two cubic factors.

The primitive roots in either factor have indices $\{1, 2, 4\}$ or $\{3, 5, 6\}$. Reversing the cyclic numbering interchanges these sets. Character projection as in Theorem 11.1 therefore shows that, after such a reversal if necessary, the roots have form

$$x_i = \eta + v_1 \zeta^i + v_2 \zeta^{2i} + v_4 \zeta^{4i}, \quad \eta = \frac{u_0}{7} = u_0 \in K.$$

On the eight-element subfield generated by ζ , the map

$$\psi(z) = v_1 z + v_2 z^2 + v_4 z^4$$

is $\mathbb{F}_2$ -linear. It is injective: a nonzero kernel element h would make $\psi(z) = \psi(z + h)$ for distinct nonzero $z, z + h$, contradicting distinctness of the seven roots. Hence $W = \psi(\mathbb{F}_8)$ has dimension three, and the roots are exactly $\eta + (W \setminus \{0\})$.

For a finite $\mathbb{F}_2$ -vector space W in a field, its monic root polynomial $L_W(Z) = \prod_{w \in W} (Z - w)$ is additive. One elementary proof is induction on the dimension. If $W = V \oplus \mathbb{F}_2 h$, then

$$L_W(Z) = L_V(Z) L_V(Z - h) = L_V(Z)^2 - L_V(h) L_V(Z),$$

which is additive whenever L_V is. Since $|W| = 8$ here,

$$L_W(Z) = Z^8 + aZ^4 + bZ^2 + dZ.$$

But $L_W(Z) = Zf(Z + \eta)$ belongs to $K[Z]$, so its displayed coefficients lie in K . Its distinct roots force $d \neq 0$. This proves (iii).

Conversely, under (iii), $ZF(Z)$ is an additive polynomial of degree eight with derivative $d \neq 0$. Its eight roots form a three-dimensional $\mathbb{F}_2$ -space W , and the nonzero roots are exactly the roots of F . For each nonzero $w \in W$, exactly three unordered pairs of distinct nonzero elements sum to w . Exactly seven unordered triples sum to zero, and exactly four sum to each nonzero w . These counts follow respectively by choosing one summand outside $\{0, w\}$, and by the seven Fano lines and the remaining 28 triples. After translation by η , pair sums are unchanged and triple sums acquire η . This proves both collisions and the identities (24).

Finally, every Galois automorphism fixes η and permutes W while preserving addition. Its action on W is faithful because the roots generate the splitting field. Thus the Galois group embeds in $\mathrm{GL}_3(\mathbb{F}_2)$. $\square$

Corollary 11.3 (A nonsolvable colliding configuration). *There are separable irreducible septics in characteristic two with colliding pair and triple sums and with Galois group of order 168, isomorphic to $\mathrm{GL}_3(\mathbb{F}_2)$.*

Proof. Let z_1, z_2, z_3 be algebraically independent over $\mathbb{F}_2$, put $L = \mathbb{F}_2(z_1, z_2, z_3)$, and let $G = \mathrm{GL}_3(\mathbb{F}_2)$ act by invertible linear substitutions. Set $K = L^G$. The fixed-field theorem for a finite group of field automorphisms gives $\mathrm{Gal}(L/K) = G$. The seven nonzero linear forms $v_1z_1 + v_2z_2 + v_3z_3$ with $v \in \mathbb{F}_2^3 \setminus \{0\}$ are distinct, form one G -orbit, and generate L . Their monic root polynomial is therefore separable irreducible of degree seven over K , with splitting field L , and its nonzero root set is an additive three-space minus zero. Theorem 11.2 applies.

The group has order $(8-1)(8-2)(8-4) = 168$. It is nonsolvable: it is generated by elementary transvections, and for distinct i, j, k each transvection $I + E_{ij}$ is the commutator of $I + E_{ik}$ and $I + E_{kj}$. Thus the nontrivial group is perfect and cannot be solvable. $\square$

The two exceptional characteristics therefore have markedly different geometry. In characteristic 29, a collision forces a single nonconstant Fourier character and a cyclic binomial equation. In characteristic two, a three-character Frobenius orbit survives and produces an affine Fano configuration, which can support a nonsolvable Galois group. The exceptional characteristic seven remains a separate unipotent problem, since its cyclic action cannot be diagonalized by seventh roots of unity.

12 A global obstruction: the universal cyclic cover is not one Kummer equation

The local pivot formulas work, but could a more ingenious single regular radical coordinate work on the entire distinct-root locus? The answer is negative in a precise algebraic sense. We prove it, rather than infer it from the zeros of a few familiar formulas.

12.1 The universal ordered configuration

Let k be algebraically closed of characteristic zero and contain a primitive p th root of unity. Put

$$A = k[x_0, \dots, x_{p-1}, \Delta^{-1}], \quad \Delta = \prod_{i < j} (x_i - x_j)^2. \quad (25)$$

Thus $\text{Spec } A$ parametrizes ordered tuples of pairwise distinct roots. Let $C_p = \langle \sigma \rangle$ act by cyclically permuting the variables, and write $B = A^{C_p}$.

No nonidentity cyclic shift fixes a geometric point: since p is prime, such a fixed point would have all coordinates equal. The quotient

$$\text{Spec } A \longrightarrow \text{Spec } B$$

is therefore a finite étale C_p -torsor. The term *torsor* means that a geometric fiber has a free transitive C_p -action; it is the algebraic version of a covering with that deck group.

A global regular Kummer presentation would be a C_p -equivariant isomorphism

$$A \cong B[T]/(T^p - a), \quad a \in B^\times, \quad (26)$$

where a generator acts by $T \mapsto \zeta^j T$ for some $j \neq 0$. The unit condition expresses the absence of ramification on the already separable locus. It is not an arbitrary strengthening: a Kummer equation with a zero radicand would fail to be étale at that zero.

The general Kummer framework distinguishes torsors given by a unit from those encoded by a line bundle whose p th tensor power is trivial [3]. We now compute the obstruction in this particular configuration explicitly.

12.2 The complete unit group

For $1 \leq a \leq (p-1)/2$ and $j \in \mathbb{F}_p$, define the oriented difference

$$\delta_{a,j} = x_j - x_{j+a}.$$

Every unordered pair occurs exactly once among these factors, up to a constant sign. Since the polynomial ring is a unique factorization domain, localizing at Δ gives

$$A^\times = \left\{ c \prod_{a=1}^{(p-1)/2} \prod_{j \in \mathbb{F}_p} \delta_{a,j}^{n_{a,j}} : c \in k^\times, n_{a,j} \in \mathbb{Z} \right\}. \quad (27)$$

The representation is unique once the chosen orientations are fixed. The cyclic action simply sends $\delta_{a,j}$ to $\delta_{a,j+1}$, without an additional sign. In multiplicative-module notation,

$$A^\times \cong k^\times \times \mathbb{Z}[C_p]^{(p-1)/2}. \quad (28)$$

Theorem 12.1 (No nontrivial eigenunit). *If $u \in A^\times$ and $\sigma(u) = \zeta^r u$, then $r = 0$ in $\mathbb{F}_p$. Consequently the universal cyclic cover has no presentation (26).*

Proof. Write u as in (27). Equality up to a nonzero constant does not change its exponents along the irreducible difference factors. Therefore $n_{a,j} = n_{a,j-1}$ for all a, j . For each a , the exponent is constant around its full cyclic orbit. The product $\prod_j \delta_{a,j}$ is itself invariant, and the constant c is fixed. Thus $\sigma(u) = u$. Since $u \neq 0$, this forces $\zeta^r = 1$.

In (26), the image of T would be a unit, because $T^p = a \in B^\times$, and would have a nontrivial eigencharacter. This is impossible by the first assertion. Even an initially unequivariant B -algebra presentation of this form would yield such an eigenunit: $\sigma(T)^p = T^p$ forces $\sigma(T)/T \in \mu_p$, and this character must be nontrivial because T generates the extension. $\square$

This rules out a single everywhere-regular Kummer coordinate on the cyclic quotient, even though rational Kummer coordinates exist over its function field. It does not rule out piecewise formulas, formulas on a dense open subset, or a nontrivial line-bundle formulation. Nor is it a claim about an unrestricted language of nested algebraic expressions with arbitrary case distinctions.

12.3 The Picard group of the cyclic quotient

Recall that $\text{Pic } B$ is the abelian group of isomorphism classes of invertible B -modules, with tensor product as its operation. Such modules are the algebraic counterparts of line bundles. Since A is a localization of a polynomial unique factorization domain, every invertible A -module is free of rank one; in particular, $\text{Pic } A = 0$.

Lemma 12.2 (Descent calculation). *For the above torsor,*

$$\text{Pic } B \cong H^1(C_p, A^\times).$$

Proof. An invertible B -module becomes free of rank one over A . Choose a generator after this base change. Transport of the generator under $g \in C_p$ is multiplication by a unit $a_g \in A^\times$. Compatibility of the action is exactly the cocycle equation $a_{gh} = a_g g(a_h)$. Changing the chosen generator changes the cocycle by a coboundary.

Conversely, a unit-valued cocycle gives a semilinear action on the free rank-one A -module. Faithfully flat descent along the finite étale torsor recovers an invertible B -module. These constructions are inverse and respect tensor products. The use of $\text{Pic } A = 0$ is precisely what permits one generator after base change for every class. $\square$

Theorem 12.3 (Picard group and the intrinsic chart obstruction). *For odd prime p ,*

$$\text{Pic}(k[x_0, \dots, x_{p-1}, \Delta^{-1}]^{C_p}) \cong \mathbb{Z}/p\mathbb{Z}. \quad (29)$$

The nonzero Fourier-character eigenspaces of A are invertible B -modules representing the nonzero powers of a generator of this group.

Proof. By (28) and Lemma 12.2, it suffices to compute first cohomology of the factors. The action on $k^\times$ is trivial, so a cocycle is just a character. Thus

$$H^1(C_p, k^\times) = \text{Hom}(C_p, k^\times) = \mu_p \cong \mathbb{Z}/p\mathbb{Z}.$$

For the additive regular module $M = \mathbb{Z}[C_p]$, a cocycle is determined by $v \in M$ satisfying

$$(1 + \sigma + \dots + \sigma^{p-1})v = 0.$$

In coordinates this says that the sum of the p coefficients of v is zero. Every such vector is a cyclic difference $(\sigma - 1)w$: solve for the coefficients of w by successive partial sums, whose consistency around the cycle is exactly the zero-sum condition. Hence $H^1(C_p, \mathbb{Z}[C_p]) = 0$. Equation (29) follows.

For each $r \in \mathbb{F}_p$, let

$$L_r = \{a \in A : \sigma(a) = \zeta^r a\}.$$

A C_p -torsor becomes the regular permutation representation after a faithfully flat base change. Its character pieces therefore have rank one, and multiplication gives isomorphisms $L_r \otimes_B L_s \cong L_{r+s}$. Descent identifies these modules with the corresponding character cocycles, up to the harmless choice of which character is called a generator. Thus L_r has class r times that generator. For $r \neq 0$, the class is nonzero and has order p . $\square$

This also explains why p th powers can be invariant without their chosen p th roots fitting into one global regular coordinate. The obstruction is not a failure of Kummer theory; it is exactly the line-bundle alternative allowed by Kummer theory.

12.4 The Fourier charts trivialize the obstruction locally

Define universal modes u_j by (14) in A , and let $a_j = u_j^p \in B$. The basic opens

$$D(a_j) \subseteq \operatorname{Spec} B, \quad j = 1, \dots, p-1,$$

cover $\operatorname{Spec} B$. Indeed, a point missing all of them lifts to a geometric point of $\operatorname{Spec} A$ with every nonconstant Fourier mode zero; inversion would make all its coordinates equal, contradicting $\Delta \neq 0$.

On $D(a_j)$, u_j is a unit and is a generator of L_j . Every other character component can be divided by the corresponding power of u_j to become invariant. Consequently

$$A[a_j^{-1}] \cong B[a_j^{-1}][T]/(T^p - a_j), \quad T \mapsto u_j. \quad (30)$$

To see that there are no missing elements, decompose an element into its p character components by averaging with powers of ζ ; each component is an invariant multiple of one of $1, u_j, \dots, u_j^{p-1}$. These powers are independent by their distinct characters, and the monic relation is exactly $u_j^p - a_j$.

The overlap formulas (18)–(19) are the concrete local gluing data for this nontrivial torsor. Theorem 12.1 says that no single chart of the same global regular kind can replace the cover. Corollary 10.2 makes the narrower coordinate-cover bound sharp even on rational examples.

13 A practical proof architecture for ProveIt

The article suggests a modular extension rather than a monolithic new formula. Existing Fourier inversion can remain unchanged. Each added module should expose the assumptions needed by its consumers and avoid a statement whose hypotheses already contain the desired factorization.

[write] Status of this section. This section is a plan. None of it is formalized, and no Lean or Rocq code was delivered with the manuscript or is shipped with this report. Two facts about the tree should be weighed before acting on it. First, descriptor separation already exists for sextics in `Algebra/PolynomialFormulas/Lean/PolynomialFormulas/SexticSeparatingInvariants.lean` (note after Theorem 7.2); the proposed `elementarySubsetDescriptor_injective` and `separatingParameter_exists` would generalize it rather than start from nothing. Second, the arbitrary-degree Fourier core has a Rocq twin, `Algebra/PolynomialFormulas/Coq/LazardGeneralFourier.v` (note in Section 1), so the atlas cluster could be built on either prover. The plan proposes Lean only.

13.1 Separating the four certificates

A coefficient-level solver should keep four logically different objects:

- (1) **Root and Galois data:** a separable irreducible polynomial, its splitting field, and a justified permutation action on its roots.
- (2) **Separation data:** an injective equivariant invariant on the relevant finite set, or a proved squarefreeness test for its orbit product.
- (3) **Invariant recovery data:** exact equalities connecting computed coefficients to $u_0, a_j, c_{j,k}$ in a specified intermediate field.
- (4) **Branch data:** one nonzero pivot and one equation $\rho^p = a_j$; coherence of all reconstructed modes then follows from Theorem 8.1.

The second item is not implied by the first. The fourth need not contain $p-1$ independent radical choices. These distinctions address two different ways an otherwise plausible solver can be unsound.

13.2 Suggested declaration-level milestones

The following are proposed names and statement boundaries, not purported existing Lean identifiers.

Proposed declaration	Essential conclusion
<code>subsetSum_injective_of_resultant_n e_zero</code>	A root p -cycle and nonzero cyclotomic resultants imply injectivity of k -subset sums.
<code>badCharacteristic_binomial_witness</code>	A common primitive cyclotomic root gives an Eisenstein binomial counterexample over a rational function field.
<code>septicTriple_badPrimes</code>	Finite determinant/orbit certificate yields exactly $\{2, 7, 29\}$.
<code>elementarySubsetDescriptor_injecti ve</code>	Subset elementary symmetric vectors distinguish subsets in every characteristic.
<code>separatingParameter_exists</code>	A union-of-root-sets argument proves the finite hitting bound.
<code>fourierPivot_ratios_fixed</code>	The pivot ratios and pivot power lie in the fixed field.
<code>fourierPivot_allBranches_shift</code>	Every branch of the single p th radical gives a cyclic shift of the original root tuple.
<code>fourierSupport_pureExtension_reali zation</code>	The rational family realizes each prescribed nonempty support.
<code>cyclicConfiguration_noEigenunit</code>	The explicit unit group has no nontrivial character unit.

The first cluster needs finite group actions, polynomial resultants, splitting fields, and finite sets. The atlas cluster needs only field algebra, finite sums, a primitive character, and a nonzero pivot. It is therefore a natural small next target, close to the inspected repository module. The global Picard calculation is conceptually stronger but needs substantially more algebraic geometry and descent infrastructure; it should not block the elementary solver improvements.

13.3 Eliminating hidden circularity

A theorem saying that a returned vector is correct *provided* it satisfies the full root factorization is not an invariant-recovery theorem. Likewise, testing numerical roots against a target polynomial is not an exact construction of the intermediate-field coefficients. A useful audit should trace every exported root theorem backward to coefficient data, nonzero tests, and explicit field operations.

For the finite classification, a checker can accept integer matrices, verify their determinants by a fraction-free algorithm, verify affine orbit coverage, and infer the prime set. It should not trust a table solely because a computer algebra system printed it. The bundled auditor is an ordinary executable model of that boundary; a verified Lean or Rocq checker would be a separate deliverable.

For the branch theorem, no polynomial coefficient expansion is necessary. The short proof works for every odd prime at once. Keeping this theorem generic avoids large quintic- or septic-specific ring-normalization goals and makes exceptional support cases explicit rather than leaving them to a generic nonvanishing assumption.

14 Further research questions and topics

The questions below are proposed follow-on research problems. Their status as previously published open problems has not been asserted. Each has a specific target and a natural first test.

Research question 14.1 (Bad characteristics in higher prime degrees). Determine $\mathcal{B}_{p,k}$ for $p = 11, 13$ and the middle subset sizes. Can the affine-orbit computation be replaced by a structural description of the prime divisors of cyclotomic norms of balanced $\{0, \pm 1\}$ vectors?

Theorem 3.3 makes this a finite exact problem with guaranteed counterexamples for every reported bad prime. A useful first deliverable is an independently checked table, followed by explanations of unusually large prime divisors. Complementation, cancellation of common subset elements, and cyclotomic automorphisms should reduce the search.

Research question 14.2 (One bad prime, different Galois groups). For higher p and $\ell \in \mathcal{B}_{p,k}$, which transitive Galois groups can occur for colliding examples? Can the characteristic-2 and characteristic-29 rigidity classifications for septics be extended to a complete classification in characteristic seven?

Theorems 11.1 and 11.2 already answer two of the septic cases: characteristic 29 forces a cyclic binomial configuration, whereas characteristic two permits a nonsolvable group of order 168. For higher primes, the number and arrangement of common cyclotomic roots should impose analogous Fourier-support constraints. Characteristic p instead requires a unipotent cyclic-operator analysis.

Research question 14.3 (Smaller deterministic separating families). For $(n, k) = (7, 3)$, replace the bound 1191 by a substantially smaller uniform number for a specified family of separating invariants. Can one exploit the Galois action to count entire collision orbits at once?

The present bound simply adds the degrees of all pair differences, often counting the same exceptional parameter many times. It is not claimed to be sharp. A good improvement must still handle finite characteristic and must distinguish the existence of one good parameter from a fixed short hitting set over the original field.

Research question 14.4 (An exact characteristic audit for matching predicates). For the additional root relations used to match invariant factors in septic formulas, determine the corresponding balanced or unbalanced coefficient vectors and their cyclotomic obstructions. Is there one finite set of excluded characteristics that certifies every matching step?

This is more precise than assuming that the prime divisors of $p(p-1)$ are the only possible problems. The characteristic-29 calculation shows that auxiliary invariants can introduce other primes. Unbalanced relations require separately treating their constant-character component.

Research question 14.5 (Minimal nonlinear covers of the cyclic configuration). What is the smallest number of regular Kummer charts needed to cover $\text{Spec } B$ when arbitrary nontrivial character sections are allowed?

We prove that one global chart is impossible and that the standard Fourier coordinate cover needs all $p-1$ of its members. These facts do not settle the optimum among nonlinear choices. This question links radical formulas to the number of generators of the nontrivial invertible modules L_j .

Research question 14.6 (Other permutation subgroups and their unit characters). Compute $\text{Pic}(A^H)$ and its radical-coordinate obstructions for other subgroups $H \leq S_n$ acting freely on ordered distinct-root configurations, beginning with the affine prime-degree groups.

The descent method still reduces the problem to $H^1(H, A^\times)$. However, permutations can reverse orientations of difference factors, so the clean cyclic splitting (28) can acquire character corrections. Those signs must be derived rather than discarded.

Research question 14.7 (Coefficient-efficient invariant ratios). For quintics and septic, construct small exact expressions for the chart data $a_j, c_{j,k}$ in a computationally convenient presentation of the abelian intermediate field, and compare them with existing large invariant formulas.

The atlas theorem guarantees that only one p th radical is necessary once these invariants are known. The unresolved algorithmic issue is computing them without already knowing an ordered root tuple. Modular reconstruction, trace pairings, and carefully chosen resolvents are plausible approaches.

Research question 14.8 (Verified finite certificates and branch reconstruction). Can the 29 norm determinants, the affine orbit partitions, and the generic one-radical chart theorem be formally verified with a small trusted kernel boundary and without native computation assumptions?

The provided certificates are deliberately small. A realistic first milestone is the septic triple classification and the one-radical theorem, followed by a coefficient-level example and a full audit of its dependency graph. This is a formalization target, not a claim that such a proof is already included.

Research question 14.9 (Numerically stable exact chart selection). Among nonzero Fourier charts, which selection criterion minimizes intermediate coefficient growth and numerical conditioning while remaining compatible with exact certification?

The algebraic theorem only distinguishes zero from nonzero. A practical hybrid solver could use approximate information to propose a pivot and exact arithmetic to certify it. The mathematical output must remain independent of the heuristic and preserve the transition identities.

15 Conclusion

Two kinds of exceptional behavior have different causes. Subset-sum collisions are arithmetic: they are controlled exactly, away from the prime degree itself, by cyclotomic norm divisibility. The exceptional characteristic 29 for septic triple sums is a concrete manifestation, with a separable irreducible cyclic witness and a complete repeated-factor resolvent. The collision locus is even rigid: characteristic 29 forces a translated pure binomial, while characteristic two forces an affine Fano configuration and an additive-polynomial quotient. A separating vector compressed to a scalar supplies a uniform repair with an explicit finite search bound.

Exceptional radical coordinates are geometric as well as arithmetic. Every nonempty Fourier support occurs even for rational examples with full affine Galois group. A nonzero pivot gives a coherent one-radical local formula, but the universal cyclic cover carries a nontrivial order- p line-bundle class and admits no single global regular Kummer generator. The right object is therefore an atlas with certified transitions, not a formula whose denominators are assumed never to vanish.

For ProveIt, the immediate gain is a sharper proof architecture: separate root separability from resolvent separation, separate invariant recovery from Fourier inversion, and derive branch coherence from one shared radical. The results here supply exact mathematical statements and small certificates for those interfaces, while leaving coefficient optimization and kernel formalization as explicit next tasks.

A Reproducibility and the finite proof boundary

A.1 Files in the archive

The archive contains this source and compiled PDF, a README, two checking programs, and generated certificate files. The principal program is `verify_results.py`; it uses exact SymPy integer and rational arithmetic, not floating-point roots. The independent program `audit_certificates.py` uses only the Python standard library to check the orbit and determinant certificates.

Typical commands are

```
python -m pip install sympy
python verify_results.py --out certificates
python audit_certificates.py --cert certificates/cyclotomic_norm_certificates.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The source uses common LaTeX packages and has no external figure or bibliography-file dependency.

[write] **Shipped names.** The commands above use the delivery layout. In this report the scripts are `code/verify_results.py` and `code/audit_certificates.py`, the certificate files are in `data/` (not `certificates/`), and the delivered PDF is not shipped. Run from the report directory, writing fresh outputs to a scratch directory so that the shipped records are never overwritten:

```
uv run --no-project --with sympy==1.14.0 python code/verify_results.py --out <tmpdir>
py code/audit_certificates.py --cert data/cyclotomic_norm_certificates.json
```

The auditor is read-only without `--out` and prints its report, which equals `data/independent_audit_report.json`. At the intake the fresh outputs of the verifier equalled the files in `data/` after removing carriage returns; the "python" field of `verification_report.json` records the interpreter and may differ.

A.2 Recorded checks

The executed full verification run used Python 3.13.5 and SymPy 1.14.0. The report `certificates/verification_report.json`² records a passing result for:

Check	Count
Exact pairwise cyclotomic resultants	850
Independent orbit-representative integer determinants	29
Finite-field counterexample families	4
Distinct repaired characteristic-29 coefficient vectors	35
Nonempty Fourier supports, degrees five and seven	78
Exact Fourier-mode identities	516
Nonzero pivot charts	224
Reconstructed root entries over all branches in those charts	10,208
Exact worked minimal polynomials	6
Solvable affine septic orbit patterns	4
Finite-field character-root patterns for septic rigidity	1,400

The Fourier test algebra is

$$\mathbb{Q}[\alpha, \zeta]/(\alpha^p - 2, \Phi_p(\zeta)).$$

²[write] Shipped in this report as `data/verification_report.json`.

Its elements are stored with exact rational coefficients in the basis $\alpha^a \zeta^b$, $0 \leq a < p$, $0 \leq b < p - 1$. The field degree calculation in Theorem 10.1 justifies this basis. The tests run through every nonempty support with coefficient one, every available pivot, all p choices of the radical branch, and all p reconstructed roots. General rational coefficients are covered by the proof, not exhaustively by this finite test.

The checks do not prove that the cited repository modules compile in a fresh environment, and no Lean or Rocq process was run for this article. Nor do these checks verify the Picard calculation: that part is the written descent proof in Section 12. Keeping these boundaries explicit prevents an exact computation from being mistaken for an unrelated formal verification claim.

B A compact dependency map

The central results have the following logical dependencies:

Cauchy theorem + cyclic operator Bézout identity
 $\implies$ safe resultant criterion,
 cyclotomic common root + Eisenstein binomial
 $\implies$ necessity at every non- p bad prime,
 finite norm tables + exact criterion + Artin–Schreier witness
 $\implies$ complete small-degree classification,
 septic norm table + characteristic-29 character projection
 $\implies$ translated-binomial rigidity,
 two cubic cyclotomic factors in characteristic two
 $\implies$ affine Fano and additive-quotient rigidity,
 subset polynomial identity + finite root bound
 $\implies$ uniform separating repair,
 Fourier covariance + one nonzero mode + inversion
 $\implies$ all-branch coherent reconstruction,
 prime-degree pure extension + geometric sums
 $\implies$ all support patterns,
 factorial configuration ring + explicit units + descent
 $\implies \text{Pic } B \cong \mathbb{Z}/p\mathbb{Z}$ and no global eigenunit.

There is no dependence of the arithmetic classification on the geometric Picard theorem, and no dependence of the Fourier chart theorem on the finite norm table. The three strands can be independently reviewed and formalized.

References

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- [2] Mohammed A. Faggal and Daniel Lazard, *Solving Quintics and Septics by Radicals*, International Journal of Scientific and Research Publications, 4(12), December 2014, pp. 1–10.

- <https://www.ijssrp.org/research-paper-1214/ijssrp-p3679.pdf>. Especially Proposition 7 and Section 5.2. The conclusions drawn in Section 6 concern separation and distinct factors, not an unqualified refutation of radical solvability.
- [3] The Stacks Project Authors, *Kummer theory*, Section 59.28, Tag 03PK. <https://stacks.math.columbia.edu/tag/03PK>. Used for the standard line-bundle interpretation of Kummer torsors; the unit and Picard-group calculations for the present configuration are proved in this article.
 - [4] Vladimir Reshetnikov and contributors, *ProveIt*, root README, inspected snapshot e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9. <https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/README.md>.
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 - [8] ProveIt project, [write] *Algebra/PolynomialFormulas/Coq/LazardGeneralFourier.v*, the Rocq counterpart of *LazardGeneralFourier.lean*. <https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/Algebra/PolynomialFormulas/Coq/LazardGeneralFourier.v>.
 - [9] ProveIt project, [write] *Algebra/PolynomialFormulas/Lean/PolynomialFormulas/SexticSeparatingInvariants.lean* and *Algebra/PolynomialFormulas/Decidability.md*. <https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/Algebra/PolynomialFormulas/Lean/PolynomialFormulas/SexticSeparatingInvariants.lean>.
 - [10] ProveIt project, [write] *Algebra/PolynomialFormulas/Coq/LazardCyclicKummerGenerator.v*. <https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/Algebra/PolynomialFormulas/Coq/LazardCyclicKummerGenerator.v>.
 - [11] ProveIt project, [write] *LazardAlternatingResolventCounterexample* and *LazardArtinSchreierRadicalCounterexample*, in *Algebra/PolynomialFormulas/Lean/PolynomialFormulas/* (.lean) and *Algebra/PolynomialFormulas/Coq/* (.v); status in *Algebra/PolynomialFormulas/LazardPaperClaimCrosswalk.md*, rows at l. 200 and 202.